

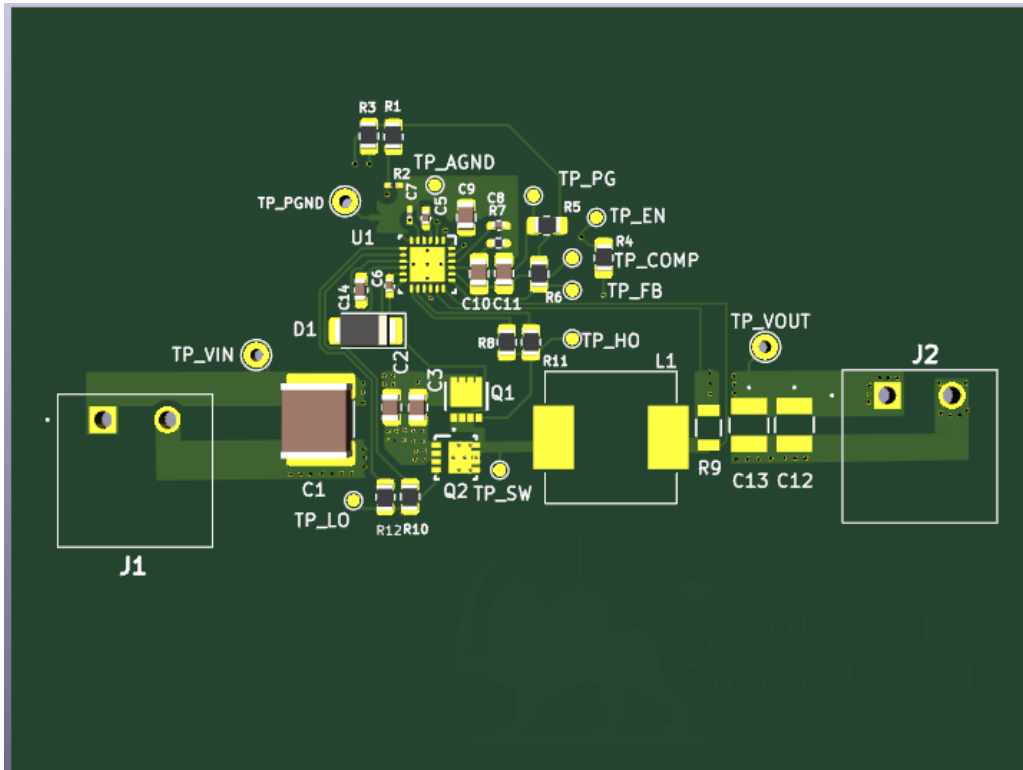
LM5141-Q1 Automotive Synchronous Buck Converter

12 V $\rightarrow$ 3.3 V, 5 A | Peak Current Mode Control

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Hardware Engineer | Power Electronics | PCB Design | LTspice | KiCad

PROJECT OVERVIEW



Project Summary

This project presents the complete design and implementation of a 12 V to 3.3 V, 5 A synchronous buck converter using the Texas Instruments LM5141-Q1 automotive-qualified controller.

The project was developed following a complete engineering workflow including system specification, power-stage design, control-loop design, LTspice simulation, schematic capture, PCB layout, and manufacturing preparation.

The design was implemented using LTspice, TI WEBENCH, and KiCad and has been released for PCB fabrication through JLCPCB.

Key Specifications

Parameter	Value
Input Voltage	8 V – 16 V
Nominal Input	12 V
Output Voltage	3.3 V
Output Current	5 A
Switching Frequency	440 kHz
Topology	Synchronous Buck
Controller	LM5141-Q1
Control Method	Peak Current Mode
PCB Stackup	4 Layers

Industry Relevance

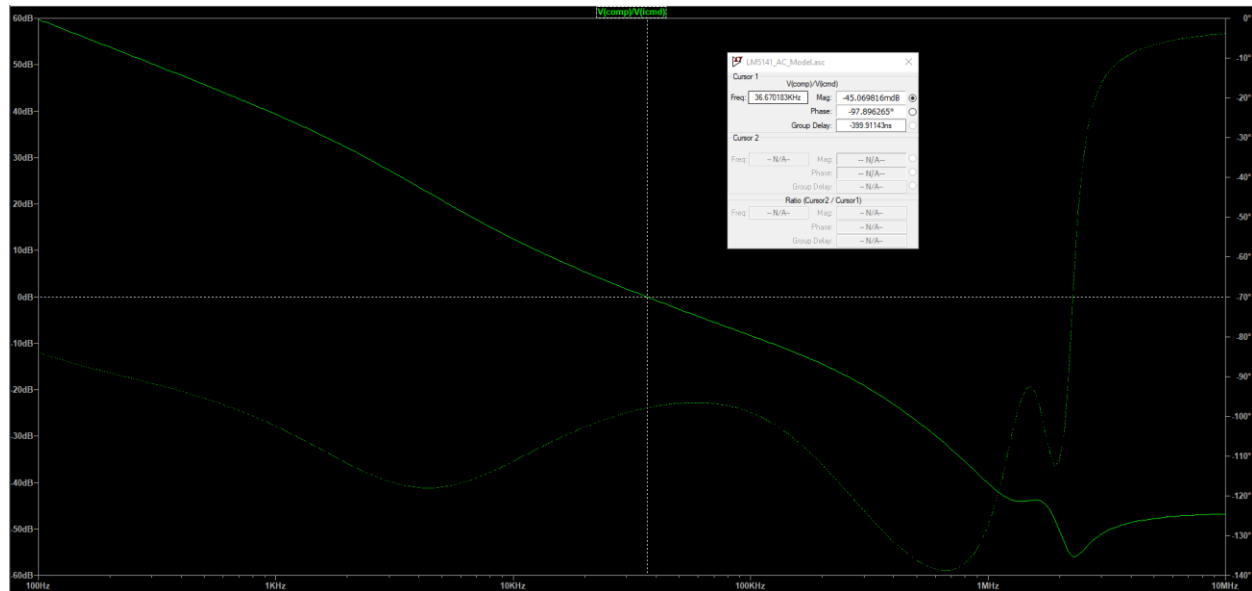
The converter was developed following automotive and industrial power electronics design practices.

Particular attention was given to EMC-aware PCB layout techniques including:

- Hot-loop minimization
- Compact switching-node geometry
- Kelvin current sensing
- AGND / PGND separation
- Optimized gate-drive routing

These design practices are commonly used in professional automotive power conversion systems and support future EMC/EMI compliance activities.

CONTROL LOOP DESIGN & SIMULATION



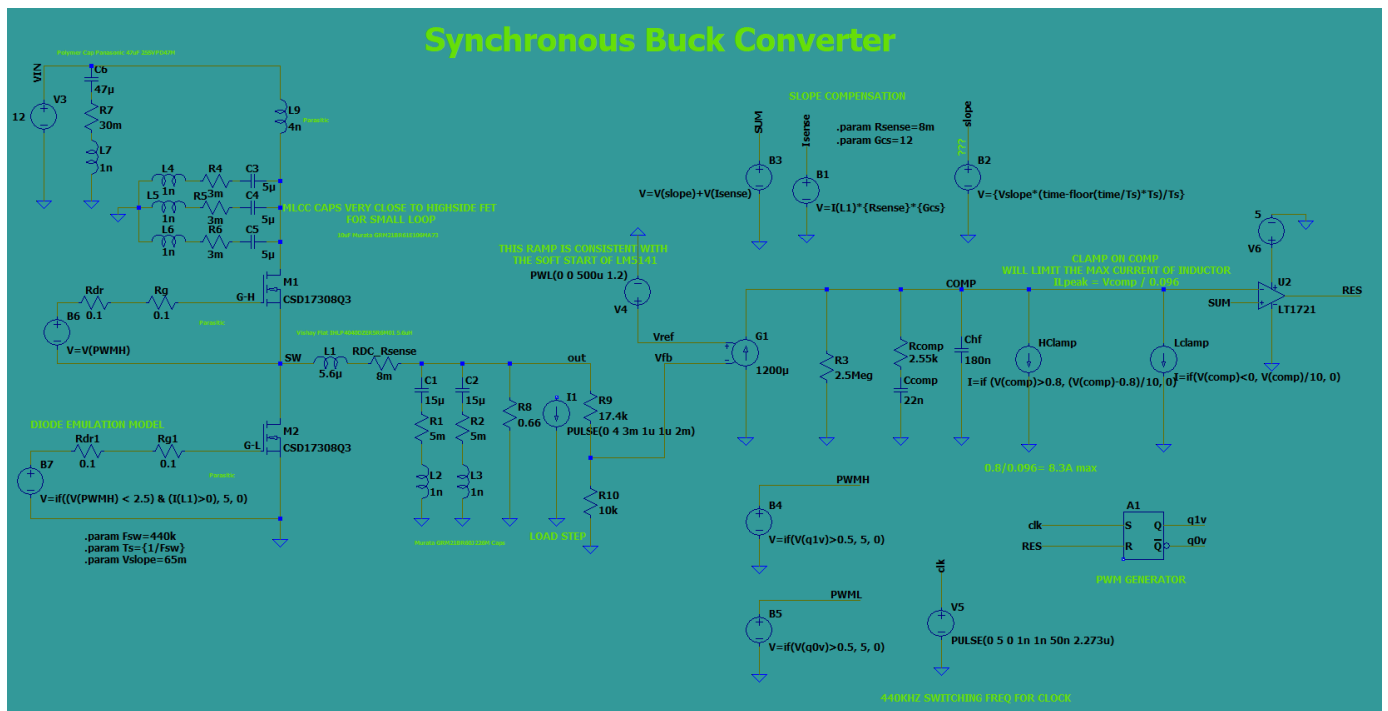
Stability Results

- Control Method: Peak Current Mode
- Compensation: Type-II
- Crossover Frequency: ~37 kHz
- Phase Margin: ~83°
- Switching Frequency: 440 kHz

The compensation network was designed and validated using LTspice averaged AC analysis. The resulting loop response provides robust stability and excellent transient performance while maintaining sufficient bandwidth for fast load regulation.

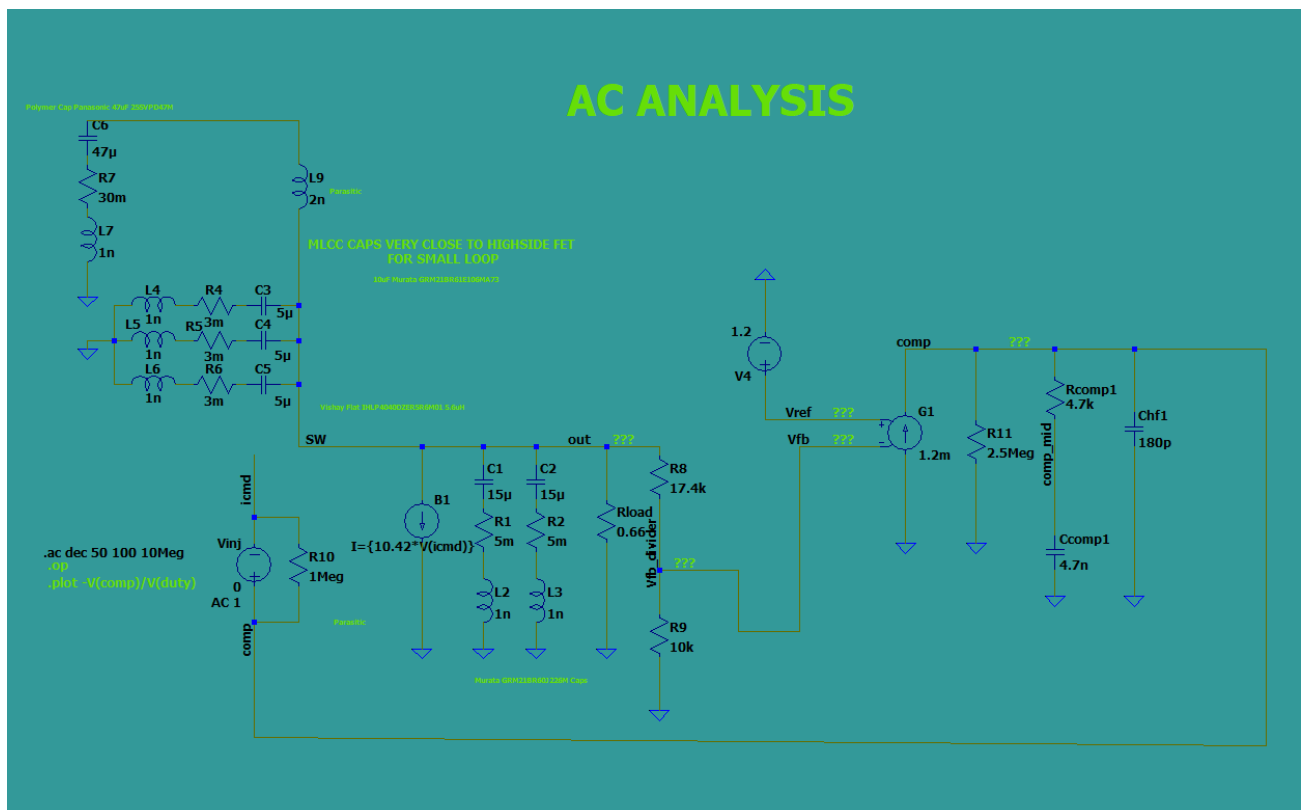
Simulation Activities

- Startup analysis
- Load transient analysis
- Line transient analysis
- Loop-gain verification
- Compensation optimization

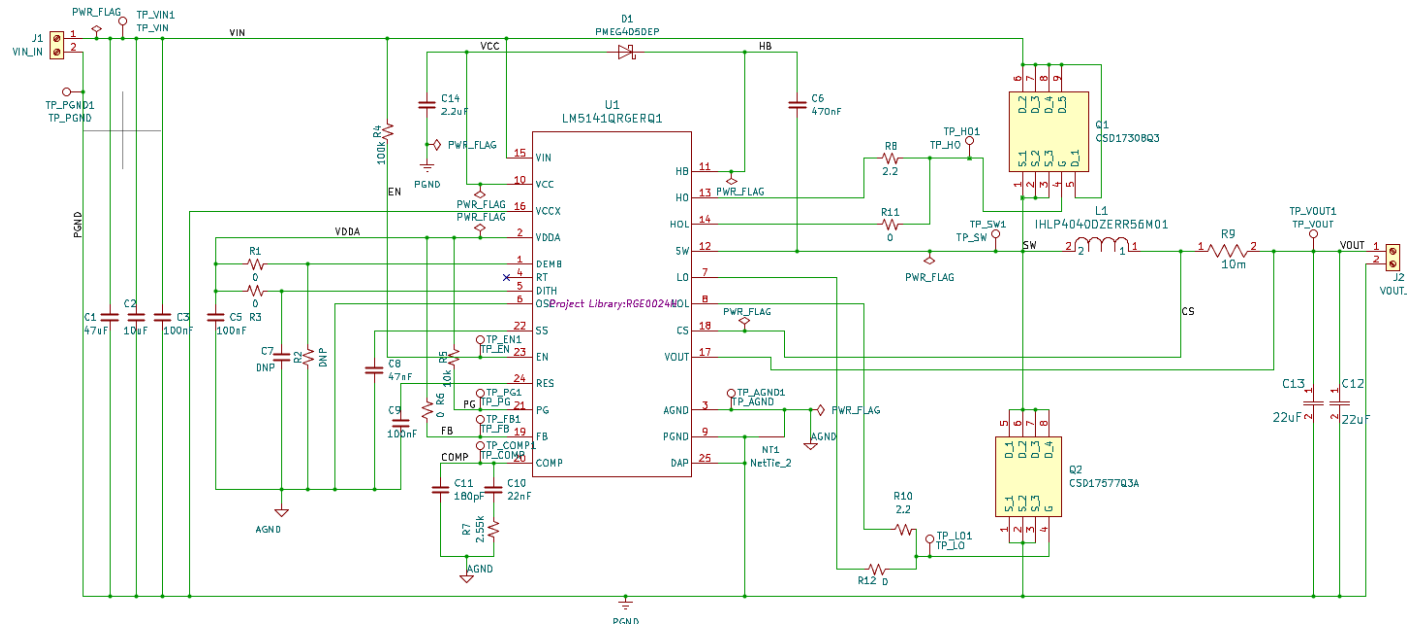


Caption:

Custom peak-current-mode control implementation including current sensing, slope compensation, PWM generation, compensation network, and soft-start functionality.



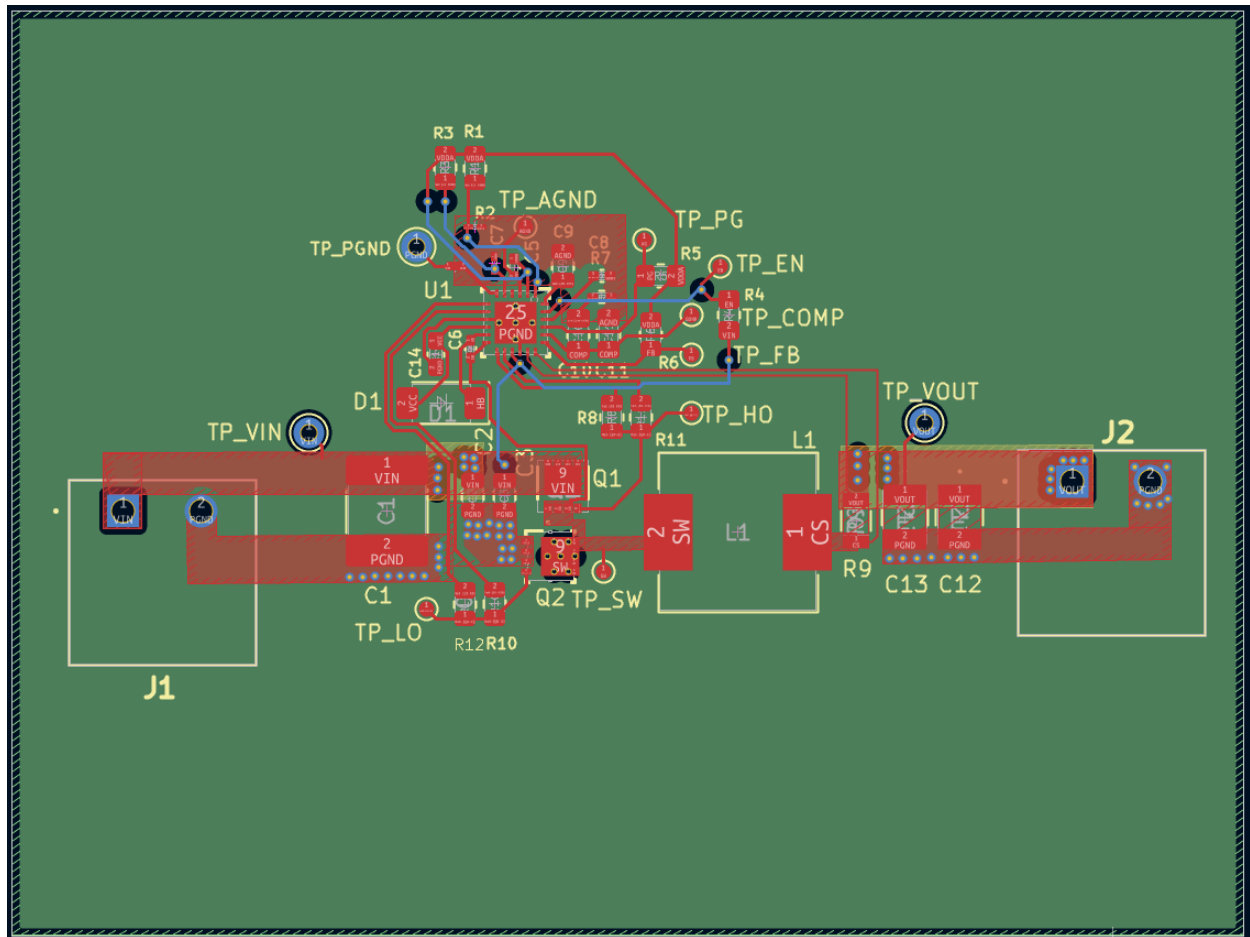
Schematic Capture



The complete converter was implemented in KiCad and includes:

- LM5141-Q1 controller
- Bootstrap circuitry
- Synchronous MOSFET stage
- Current sensing network
- Compensation network
- Input and output filtering

PCB Layout



The PCB was implemented as a four-layer design using industry-standard switching power supply layout techniques.

Layout Highlights

- Hot-loop minimization
- Compact power-stage placement
- Kelvin current sensing
- AGND / PGND separation using net tie
- Short gate-drive loops
- Test-point strategy for validation
- Manufacturing-ready layout

Engineering Skills Demonstrated:

Power Electronics Design - DC-DC Converter Design - Peak Current Mode Control - Compensation Design - Loop Stability Analysis - LTspice Simulation - PCB Design (KiCad) - EMC-Aware Layout Techniques - Design for Manufacturing (DFM) - Technical Documentation

Engineering Tools:

- ✓ LTspice
- ✓ KiCad
- ✓ TI WEBENCH
- ✓ JLCPCB
- ✓ Datasheet Analysis
- ✓ PCB Design

Repository

GitHub Repository:

<https://github.com/farhad-ekramitalab/LM5141-Synchronous-Buck-Converter>

Future Validation Plan

The next phase of the project includes hardware assembly and laboratory characterization. Planned validation activities include:

- Startup verification
- Output ripple measurement
- Load transient testing
- Efficiency characterization
- Thermal analysis
- EMI/EMC-oriented waveform evaluation

This project demonstrates the complete workflow of a modern power electronics design from specification through simulation, PCB implementation, and manufacturing release.